

		object based on N's 1 st and 2 nd law.
8	B	For elastic collision, relative speed of approach ($u_1 - u_2$) before collision is equal to the relative speed of separation ($v_2 - v_1$) after collision.

Qn	Answer	Solution
		$u_1 - u_2 = 5 - (-2) = 7$ $v_2 - v_1 = 7 \text{ m s}^{-1}$ Option B and D are possible answers. Option D is incorrect as it is not possible for the large mass to still continues in the same direction at 5 m s^{-1} after collision. Checking total K.E. before and after collision, option B is the only answer.
A	B	Increase in elastic potential energy is area under the force-length graph for F_s to